AFSar: An Anchor-Free SAR Target Detection Algorithm Based on Multiscale Enhancement Representation Learning

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***Abstract*— Unlike optical images, synthetic aperture radar (SAR) images have unique characteristics, such as few samples, strong scattering, sparseness, multiple scales, complex interference and background, and inconspicuous target edge contour information. Current SAR target detection algorithms have difficulty in balancing accuracy and speed, and the performance of these algorithms is relatively limited, thus making it difficult to deploy practical applications. To this end, this article proposes AFSar, an innovative anchor-free SAR target detection algorithm based on multiscale enhancement representation learning. First, we introduce the latest anchor-free architecture YOLOX as the basic framework. Second, to reduce the computational complexity of the model and to improve the ability of multiscale feature extraction, we redesigned the lightweight backbone, namely, MobileNetV2S. Furthermore, we propose an attention enhancement PAN module, called CSEMPAN, which highlights the unique strong scattering characteristics of SAR targets by integrating channel and spatial attention mechanisms. Finally, in view of the multiscale and strong sparse characteristics of SAR targets, we propose a new target detection head, namely, ESPHead. ESPHead extracts the features of targets with different scales by using dilated convolution with different dilated rates, so as to enhance the detection ability of the model for targets with different scales. The results of ablation experiments on the SSDD dataset show that the mAP of our algorithm reaches 0.977, while the Flops is only 9.86 G, achieving state of the art.**

***Index Terms*— Anchor-free, deep learning, enhancement representation learning, multiscale learning, synthetic aperture radar (SAR) target detection.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor. SAR has all-weather characteristics and can maintain high observation ability under harsh environments. With the development of SAR imaging technology [9], [12], SAR is widely used in military and civilian fields. For example, military applications include military reconnaissance, area surveillance, precision strikes, damage assessment, and nuclear explosion detection. Civilian applications include agricultural planning, forestry monitoring [16], water conservancy survey, urban management, environmental protection, geological and mineral prospecting, ecological environment, and natural disaster prediction. SAR target detection aims to locate the position and target class of the target in the image. As an important research direction, SAR target detection has received wide attention.

Traditional SAR target detection is mainly based on manually designing features method and the constant false alarm rate (CFAR) [19]. Methods based on manually designed features include GLRT [21], SEPT [22], and ARI-FG [23]. Because they heavily rely on manually designed features, these perform well in detecting certain types of targets. However, with the change of target characteristics and the image has defocus problem, the detection performance drops sharply, and the generalization performance is poor. Methods based on the CFAR include multiscale CFAR [24], superpixel two-level CFAR [25], and CSM [26]. CFAR relies on clutter statistics, as the SAR image resolution becomes larger and the background becomes increasingly complex, the distribution parameters of the target may be estimated incorrectly, leading to a serious decline in detection performance.

Due to the wind and waves at sea, the defocus problem may occur when imaging ships at sea [27]. The traditional CFAR determines a threshold after processing the input image and compares this threshold with the image signal. If the input image exceeds this threshold, it is judged as having a target; otherwise, it is judged as having no target. Due to the defocus problem in SAR imaging, it may affect the threshold in CFAR processing and the detection accuracy. In recent years, with the development of deep learning, methods based on convolutional

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neural networks automatically learn the general characteristics of targets from numerous data through the continuous stacking of convolutional layers, avoid SAR image defocus problem, and enhance the generalization performance of algorithms. Many scholars have proposed a series of detection models based on convolutional neural networks. Currently, target detection methods based on deep learning can be divided into three categories: methods based on regional suggestions, methods based on target regression, and methods based on anchor-free architecture.

The detection algorithm based on region proposals are usually divided into two stages. In the first stage, a single module generates some region candidate proposals, and the second stage classifies and regresses them. The classical detection algorithms based on region proposals are R-CNN [28], Fast R-CNN [29], Faster R-CNN [10], and so on. The two-stage detection algorithm performs better but the speed is slower.

The detection algorithm based on target regression treats the detection problem as a regression problem. The detector learns the target class and position regression according to the extracted features. Classic target regression-based detection algorithms include the SSD [7], RetinaNet [4] and, YOLO [1], [30], [31] series. Because target regression algorithms reduce the extraction of region candidate box, they directly carry out the regression and classification of the target; consequently, these algorithms are faster, but their detection performance is poor.

Anchor-free methods first detect the points at the upper left and lower right corners of the target and then form a detection frame through the combination of corner points or directly detect the central area and boundary information of the object to remove the predefined anchor frame. These algorithms avoid memory usage in the complex operation and training of the anchor mechanism. Anchor-free methods are still developing rapidly, and there are several main detection frameworks, such as CenterNet [18], FCOS [15], and YOLOX [32].

Most target detection algorithms are developed according to natural images. Due to the scattering characteristics and sparse target characteristics of SAR images, the performance of algorithms based on optical images directly applied to the SAR field is poor. Therefore, researchers have developed many SAR target detection algorithms based on deep learning according to the characteristics of SAR images. Currently, the ones with better performance are CenterNet++ [17], R2FA-Det [33], FBR-Net [3], and so on.

However, detailed investigation reveals that current SAR target detection tasks still have the following problems, which are worthy of further discussion.

1) Most current CNN-based SAR target detection algorithms [34]–[38] directly migrate the mature target detection network in the optical image field to the SAR target detection task without specifically considering the characteristics of the SAR image to implement a targeted architecture design.
2) In the pursuit of higher accuracy, most studies ignore the model's high Flops, which leads to many model parameters and a slow real-time inference speed. Therefore, it is difficult to deploy these models to edge devices (such as airborne/spaceborne/missile-borne military reconnaissance equipment or civilian equipment) for practical application, thus greatly limiting the research value of the model.
3) Anchor algorithms are the mainstream in current research. However, SAR images are very sensitive to the size and aspect ratio of the anchor due to the large imaging angle, high resolution, and variable target scale. It takes considerable time and effort to adjust hyperparameters of the anchor. Manual parameter adjustment experience greatly affects the performance of the model, thereby resulting in unstable performance. In addition, anchor-based methods need to generate many candidate anchor, thereby greatly increasing the calculation of the model; anchor-based methods use the target contour and texture information as a learning guide. However, the SAR target itself exhibits extremely strong scattering and sparsity. In most cases, the edge contour information of the target is insufficiently obvious. This leads to the use of such methods for target feature learning, and it is usually difficult to achieve good results.

Therefore, in response to the abovementioned key difficulties and abandoning the inherent defects of the anchor-based architecture, this article proposes AFSar, an innovative anchor-free SAR target detection algorithm based on multiscale enhancement representation learning. Our method balances the accuracy and speed and achieves the state-of-the-art performance on the SSDD dataset.

The main contributions of our work can be summarized as follows.

1) As current anchor-based SAR target detection algorithms have difficulty solving the problems caused by the multiscale, strong scattering, and sparsity characteristics of SAR targets, we propose an anchor-free algorithm. We introduced the latest anchor-free architecture YOLOX as the basic framework and then carried out a series of optimization designs based on the characteristics of the SAR target. The introduction of the anchor-free architecture better caters to the unique characteristics of the SAR target and avoids a series of hyperparameter settings and architecture design related to the anchor, thus reducing the calculation introduced by the anchor.
2) To adapt to the multiscale and few-sample characteristics of the SAR target, to meet practical requirements, to further reduce complexity, and to improve the real-time inference speed, we restructured the backbone from the lightweight perspective. We first introduced MobileNetV2 [39] as the basic network for feature extraction; furthermore, we implemented clever component splitting, combination, and stacking on the bottleneck module to build a new backbone, referred to as MobileNetV2S. Our goal is to extract and retain the multiscale salient feature information of the SAR target as efficiently as possible by reducing the computational complexity of the model.
3) To highlight the significant and strong scattering feature information of targets, we propose an attention

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enhancement PAN module, called CSEMPAN. This module further enhances the feature aggregation and extraction ability through fusion channel and spatial attention, thereby highlighting SAR target features and suppressing background information.

4) Aiming at the multiscale and strong sparse characteristics of SAR targets, to extract the salient information of different scales, we propose a new detection head, namely, ESPHead. Convolution modules with different dilated rates are constructed to obtain receptive fields of different scales and extract spatial hierarchical feature information; furthermore, the feature is merged through the add operation to increase the network's ability to locate the targets of different scales.

II. Related Work

A. Optical Image Target Detection

Object detection as a basic computer vision task, with the development of deep learning, and object detection has also ushered in a period of rapid development. At present, the mainstream optical image detection algorithms mainly include anchor-based and anchor-free algorithms.

1) Anchor-Based Detection Algorithm: The algorithm anchor-based can be divided into two stages and one stage algorithm. R-CNN [28] is the first detection algorithm that successfully applies CNN. R-CNN through selective search method generates 2000 candidate regions, and each candidate region is resized to a fixed size, then through CNN to extract features of each candidate. Since many candidates generate the same area, extracting features for each candidate generates causes a lot of repeated calculations. To solve this problem, Girshick [29] proposed Fast R-CNN in 2015. Fast R-CNN directly extracts the features of the whole image instead of each candidate box, thereby avoiding repeated calculations. But the candidate box of Fast R-CNN is also generated by selective search, it is very time-consuming. In 2015, Ren *et al.* [10] proposed Faster R-CNN to solve the overly slow speed of the selective search. Faster R-CNN uses the region proposal network (RPN) to generate candidate regions. Such regions greatly improved detection speed and accuracy.

The SSD [7] is a one-stage algorithm, it generates multiple default boxes at each feature point in the feature graph through a feature extraction network, and the algorithm predicts four coordinate values (x, y, ω, h) of class confidence and regression for each default box. YOLO [1], [30], [31], series is another one-stage algorithms, and it is divided into input images S $\times$ S grids. Each grid is responsible for detecting the objects in the grid on which the truth target falls through the center point of the downsampled target. The principle of RetinaNet is similar to that of the YOLO series except that a new loss, namely, focal loss, has been designed to solve the foreground and background imbalance that occurs when predicting.

2) Anchor-Free Detection Algorithm: Anchor-based algorithms involve many hyperparameters. The setting of hyperparameters greatly affects the final result, this is very inconvenient for inexperienced researcher. CenterNet [18] simplifies the detection problem to a standard critical point assessment problem by regressing the location, size, and other attributes of the object from its center point. FCOS [15] maps every point in the feature map back to the original image through the feature extraction network. If the point in the feature map is mapped back to the original input point within the corresponding GT, the point will be used as the positive sample of training for target regression. This pixel-by-pixel regression method greatly improves the recall rate. However, pixel-by-pixel regression also introduces many redundant positive samples. YOLOX [32] considers the central 3 $\times$ 3 region as a positive sample, and it is possible to match two positive samples of GT with the SimOTA matching rule to optimize the match to the most appropriate sample.

B. SAR Image Target Detection

1) Traditional SAR Image Target Detection: The CFAR method is one of the most commonly used techniques for traditional SAR target detection. Many researchers have developed algorithms based on CFAR. Leng *et al.* [40] proposed a bilateral CFAR algorithm by combining the intensity distribution and spatial distribution of SAR images to reduce the interference of ocean clutter in SAR images. Pappas *et al.* [41] proposed a superpixel-level CFAR detector algorithm, which uses SPs instead of rectangular sliding windows to define the guard bands and background of the CFAR to reduce the false detection rate. Wang *et al.* [42] fused the correlation between pixel intensity and pixels into an IS index and proposed an IS domain CFAR method.

2) SAR Image Target Detection Based on Deep Learning:

a) Anchor-based SAR image detection method: Because the imaging mechanisms of SAR images and optical images are very different, many scholars have developed many SAR image detection algorithms based on optical image detection algorithms. Cui *et al.* [37] combined a dense attention pyramid network with the attention module CBAM which proposed a multiscale ship detection method. Chen *et al.* [33] designed a light nonlocal attention module of magnitude and embedded it in an FPN. The attention module not only enhances the features related to the object but also fully suppresses background interference. Li *et al.* [43], by exploiting the spatial and frequency-domain complementary features, developed a multidimensional domain for SAR ship detection.

b) Anchor-free SAR image detection method: Because anchors rely too much on parameter adjustment, in recent years, many researchers have proposed a series of algorithms based on anchor-free. Gao *et al.* [44] combined a lightweight feature extractor and feature aggregation network based on a center point ship predictor (CSP) and proposed an anchor-free detector. Sun *et al.* [45], on the basis of FCOS, proposed a class-position (CP) module to optimize the position regression branch features and redesigned target classification and bounding box regression methods to optimize training. Guo *et al.* [17] proposed a CenterNet++ network. This model mainly consists of a feature refinement module, feature pyramid fusion module, and head enhancement module.

III. Methods

Aiming at the key difficulties and combined with the unique imaging characteristics of SAR targets, this article abandons

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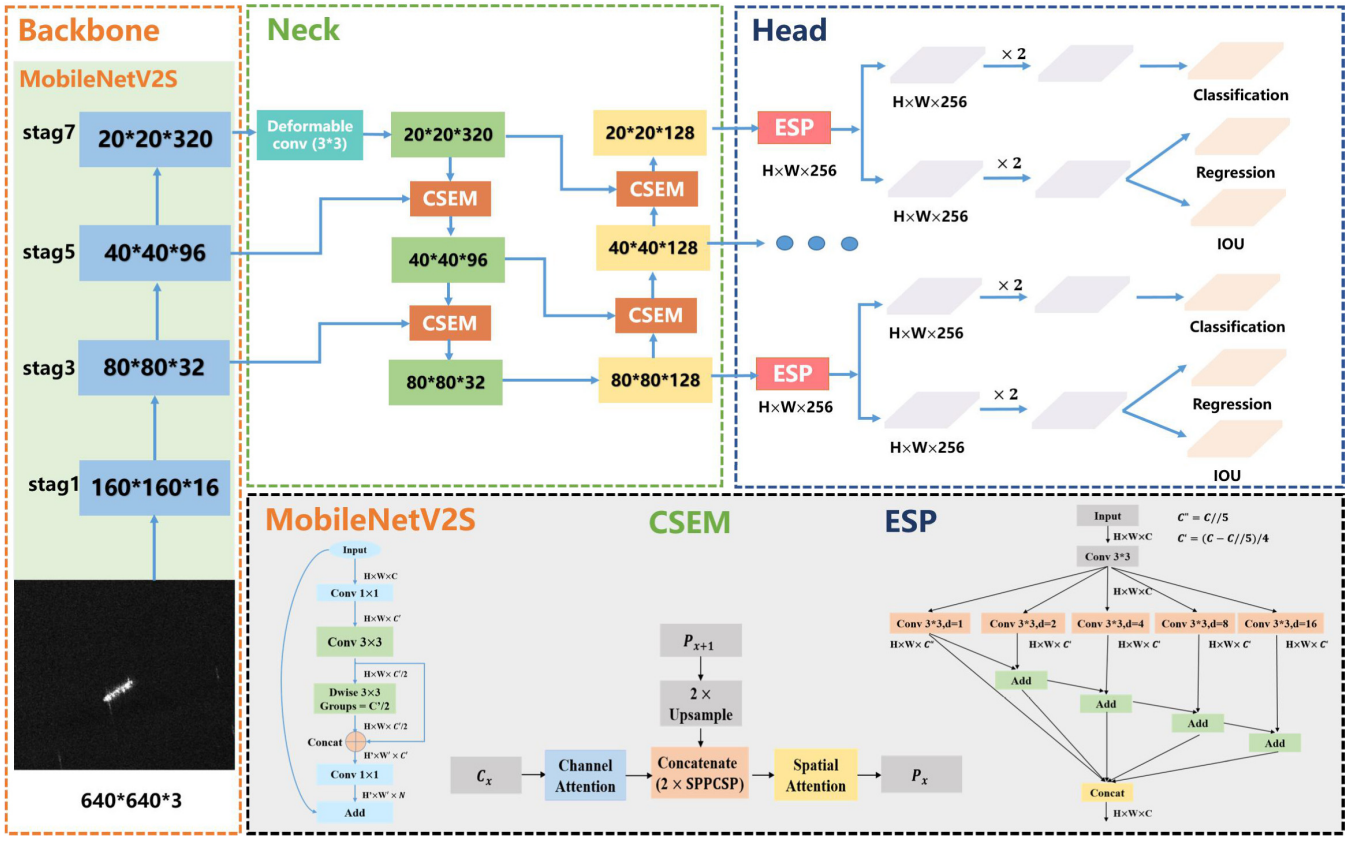


Fig. 1. Overall architecture of the AFSar network.

the inherent shortcomings of the anchor-based architecture and proposes AFSar, which is an innovative anchor-free object detection algorithm based on multiscale enhancement representation learning. In view of the existing anchor-based SAR target detection algorithm, which has difficulty solving the problems caused by SAR image target characteristics, such as multiple scales, strong scattering and sparseness. First, we introduced the anchor-free architecture and the latest anchor-free YOLOX as the basic framework. Second, to reduce the complexity of the model, we redesigned the backbone (MobileNetV2S). Furthermore, for the strong scattering characteristics of the SAR target, to highlight the salient feature information of SAR targets, we propose an attention enhancement PAN module (CSEMPAN) to increase the classification and positioning accuracy. For the multiscale and strong sparse characteristics of SAR targets, significant information on SAR targets of different scales is extracted, and a new target detection head (ESPHead) is designed.

A. Overall Architecture

The overall framework of our proposed AFSar is shown in Fig. 1. AFSar includes three main modules: backbone-MobileNetV2S, CSEMPAN, and ESPHead. The overall process is that after feature extraction by MobileNetV2S, feature maps of three scales are output to CSEMPAN. CSEMPAN conveys strong semantic features of the image from the top down and strong localization features of the target from the bottom up and aggregates the parameters of different detection layers to fully extract features. Finally, ESPHead is used to identify the target location and target class from the input features.

First, we introduce the newest anchor-free target detection framework, YOLOX, as the basic architecture. As an anchor-free detection algorithm proposed recently, YOLOX avoids the need to set anchor parameters. Furthermore, considering the particularity of SAR images, SAR images are difficult to obtain. Currently, the number of open-source SAR image datasets is small. To prevent overfitting, we proposed an improved lightweight backbone MobileNetV2S as the feature extraction network. Furthermore, after the feature extraction of the SAR image, the attention-enhanced PAN uses features of different scales for parameter aggregation to better extract the multiscale information of the objects. Finally, after the attention-enhanced PAN, ESPHead first extracts the spatial hierarchical information of different receptive fields through different dilation rates in the ESP module and finally uses the decoupled head to predict the target class information and position regression information. The postprocessing uses the SimOTA algorithm to match the positive and negative samples to calculate the loss. The feature extraction and downsampling operation determines the center point of the target truth box GT on the feature map. SimOTA considers the center 3×3 area to be a positive sample; that is, the positive samples of each GT are expanded from 1 to 9, then SimOTA calculates the IOU of the positive sample and GT and calculates the loss to determine the dynamic_k of each GT. For each GT, the top dynamic_k anchors with the smallest loss ranking are taken as positive samples, and the rest are negative samples. The loss is calculated to supervise model training.

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TABLE I
ARCHITECTURE OF MOBILENETV2

Stage	Input	output	Operator	t	c	n	s
	$224^2 \times 3$	$112^2 \times 32$	Conv2d	-	32	1	2
Stage1	$112^2 \times 32$	$112^2 \times 16$	Bottleneck	1	16	1	1
Stage2	$112^2 \times 16$	$56^2 \times 24$	Bottleneck	6	24	2	2
Stage3	$56^2 \times 24$	$28^2 \times 32$	Bottleneck	6	32	3	2
Stage4	$28^2 \times 32$	$14^2 \times 64$	Bottleneck	6	64	4	2
Stage5	$14^2 \times 64$	$14^2 \times 96$	Bottleneck	6	96	3	1
Stage6	$14^2 \times 96$	$7^2 \times 160$	Bottleneck	6	160	3	2
Stage7	$7^2 \times 160$	$7^2 \times 320$	Bottleneck	6	320	1	1

B. Lightweight Backbone-MobileNetV2S

YOLOX backbones use large-scale network architectures, such as DarkNet53 and CSPNet. Such backbones may have high detection accuracy, but due to the large size of the model, their inference speed is slow. Therefore, from the perspective of practical application, in this article, we design a lightweight backbone network for the characteristics of SAR targets. As a lightweight feature extraction backbone, MobileNetV2 has been widely used in a variety of computer vision tasks since its inception. After our evaluation and considering the few and multiscale characteristics of the SAR image dataset, we use MobileNetV2 as the basic network. Further improvements are made to the backbone network; the improved version of the network is referred to as MobileNetV2S. In contrast to the usual classification process when MobileNetV2 is used in target detection, we divide the classification into seven stages (as shown in Table I), where t is the expansion factor, C is the number of channels of the output feature map, n is the number of bottlenecks, and s is the step size of the convolution operation. When the step length of the first convolutional layer of the bottleneck of each stage is 1, the residual connection is used; otherwise, the residual connection method is adopted. Each bottleneck in MobileNetV2 contains three parts. First, the expansion convolution layer with a convolution kernel size of 1×1 is used to expand the number of channels of the feature map, and second, the convolution kernel size is a 3×3 depthwise separable convolution learning image, and the number of channels of the image is compressed. Finally, the projection convolution layer with a convolution kernel size of 1×1 is used to expand the number of channels of the feature map again. This inverted residual structure combined with a deeply separable convolution operation greatly reduces the number of model parameters.

Although the depthwise separable convolution greatly reduces the number of model parameters, the feature capture ability of the depthwise separable convolution is usually weaker than that of the ordinary convolution. Inspired by GhostNet [46], as shown in Fig. 2, we divide the depthwise separable convolution into two steps. The convolution is replaced with a normal convolution and a depthwise separable convolution. The feature map is reduced to half of the original number of channels through ordinary convolution, the depthwise separable convolution further extracts

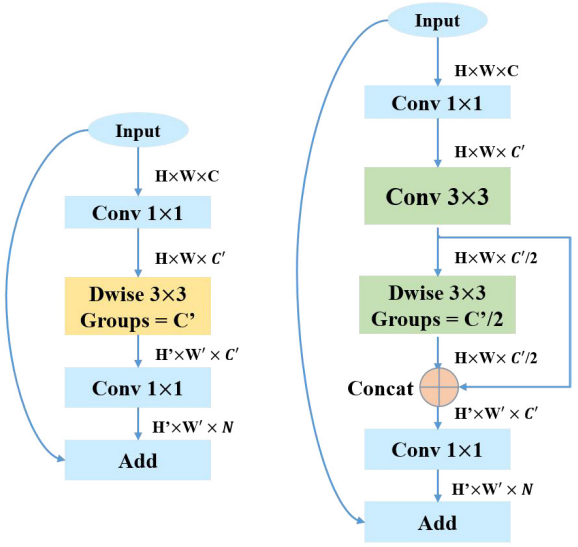


Fig. 2. Bottleneck of MobileNetV2S. On the left is MobileNetV2, and on the right is MobileNetV2S.

the features of the ordinary convolution input, and finally, both are joined together. Compared with using only a set of depthwise separable convolution, our operation can obtain the semantic information of the data, thereby further improving the modeling ability of the model and increasing the accuracy of the classification and positioning.

C. Attention Enhancement PAN-CSEMPAN

Considering the mechanism characteristics of CNN, the more convolutional layers are stacked, the larger the receptive field of feature maps is, and the more semantic information and less spatial information are contained in feature maps. The classification of targets mainly depends on semantic information, and the localization of targets depends on the receptive field of feature maps. The receptive field of shallow feature maps is smaller for the localization of small targets, and the receptive field of deep feature maps is larger for the localization of large targets. To balance the classification and localization, a top-down and bottom-up multiscale feature aggregation strategy was adopted to fuse shallow and deep feature map information. Due to the scattering characteristics of SAR image imaging, the distinction between the target and background is not high, and the traditional PAN does not

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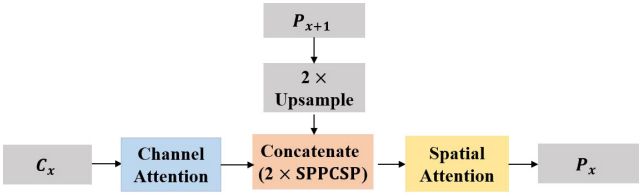


Fig. 3. CSEM module structure.

consider the significant information extracted from the target, thus resulting in a high probability of the background being detected as a target.

To solve the above problems, we designed an attention enhancement PAN module, referred to as CSEMPAN, whose architecture is shown in the neck of Fig. 1. As this figure shows, the core module in the AFSar network is a new hybrid attention enhancement module that we have built, called CSEM, which is used to suppress background information and highlights the saliency of the target. The structure of CSEM is shown in Fig. 3. The CSEM module can be divided into three parts: channel attention, SPPCSP, and spatial attention. The working principle of CSEM can be summarized as follows:

$$p_{x+1} = \text{conv}(c_{x+1}) \tag{1}$$

$$c_{x_s} = \text{concat}(\text{CA}(c_x), \text{Upsample}(p_{x+1})) \tag{2}$$

$$c_{x_f} = \text{sppcsp}\big(\text{sppcsp}\big(c_{x_s}\big)\big) \tag{3}$$

$$p_x = \text{SA}\big(c_{x_f}\big) \tag{4}$$

where CA denotes channel attention and SA denotes spatial attention.

The feature C_x extracted by the backbone network first passes the channel attention, and the different channels of the input feature map are differentiated and weighted to highlight the different importance of the channels. Considering the size of the SAR image dataset, to prevent overfitting, we choose the most commonly used and least parameter SE [47] module as the channel attention enhancement feature extraction. As shown in Fig. 4, the SE module is divided into three main steps: squeeze, excitation, and scale. The input feature maps the feature map to a size of $1 \times 1 \times c$ through the squeeze operation (c is the number of channels), and then the excitation operation uses two convolution kernels with a size of 1×1 to learn the weight coefficients of each channel. Finally, the scale exponentially multiplies the weights for each channel (between 0 and 1) by the original feature. On the basis of SE, we add the original features, add them after SE attention, and finally convolve them again with a 1×1 convolution layer. The working principle of the channel attention module can be summarized as follows:

$$z_c = F_{\text{sq}}(u_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i, j) \tag{5}$$

$$s_c = F_{\text{ex}}(z, W) = \sigma\left(B\left(W_2 B\left(\delta\left(W_1 z\right)\right)\right)\right) \tag{6}$$

$$\tilde{x} = \text{conv}(s_c \cdot u_c + u_c). \tag{7}$$

The resolution of the lower layer feature C_{x+1} is twice as small as that of C_x. Before the features are concatenated, the feature of C_{x+1} needs to be upsampled twice. After upsample, the features C_x and C_{x+1} phase concatenate, and the features after concatenation are extracted by SPPCSP twice. As shown in Fig. 5, SPPCSP has one more SPP module than the common CSP module, that is, three maxpool operations with the size of convolution kernels. Since there are targets of different sizes in SAR images, the SPP module is introduced to extract spatial feature information of targets of different sizes through maxpool of three different convolution kernels to adapt to multiscale target information.

The spatial information of the feature directly determines whether the positioning of the target is accurate. To suppress the background information and extract the target information significantly, we add a spatial attention module after the SPPCSP operation to highlight the spatial information of the target. As shown in Fig. 6, the spatial attention module first divides the input features into g groups through grouping convolution (we set g to 8), and after grouping, each group maps the feature map to 1×1 size and sums each group by global average pooling. The dot product operation is performed on the group feature map. After the dot product operation, the feature is summed along the channel dimension to obtain a spatial attention mask of the same size as the original feature map, and the number of channels is 1. The spatial attention mask uses the sigmoid operation to divide the space. The weight of each pixel above is mapped to 0–1 and multiplied with the original feature map to highlight the salient features of the target. Finally, the g group features are respliced to the size of the original feature map through the concat operation. The working principle of spatial attention can be summarized as follows:

$$x_1, x_2, \ldots, x_g = \text{split}(x) \tag{8}$$

$$n_i = F_{\text{GAP}}(x_i) = \frac{1}{m} \sum_{i=1}^{m} x_i \tag{9}$$

$$c_i = \text{sum}(n_i \cdot x_i) \tag{10}$$

$$x = \text{concat}(c_i \cdot x_i) \quad i = 1, 2, \ldots, g. \tag{11}$$

In (8)–(11), the grouped convolution represents global average pooling, is the summation operation along the channel direction, and represents splicing.

D. ESPHead

Generally, the resolution of the SAR image is relatively large, and the range of the length and width of the target is relatively large, thus resulting in a large difference in the spatial information of the target on the feature map. To obtain the information on different scale targets, we added an ESP module before the feature enters the head. As shown in Fig. 7, the feature input to the ESP module first uses a convolution kernel with a size of 3×3 to further extract the features and then uses the convolution kernel with the same size (3×3), but the dilated rates are 1, 2, 4, 8, and 16. The convolution changes the convolution's reception field to obtain the target's multiscale spatial information. Since the number of channels is not necessarily a multiple of 5, we divide the number of channels by 5 and round up and obtain the feature with the number of channels C'' as the convolution output with a dilated rate of 1, and the remaining features

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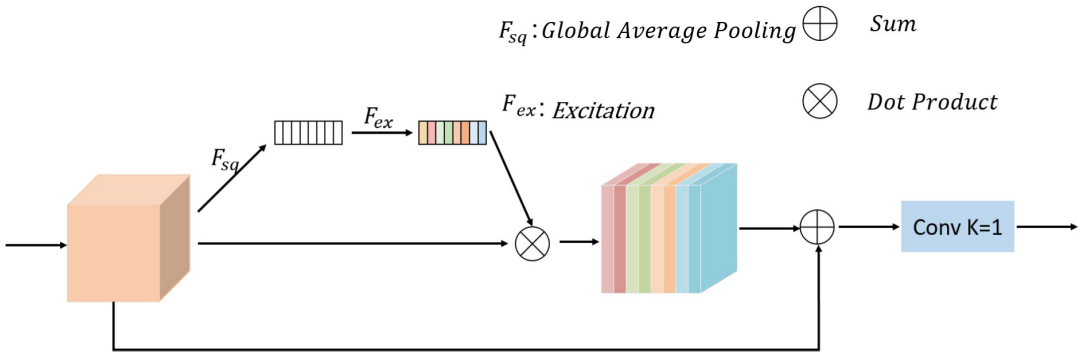


Fig. 4. Structure of channel attention module.

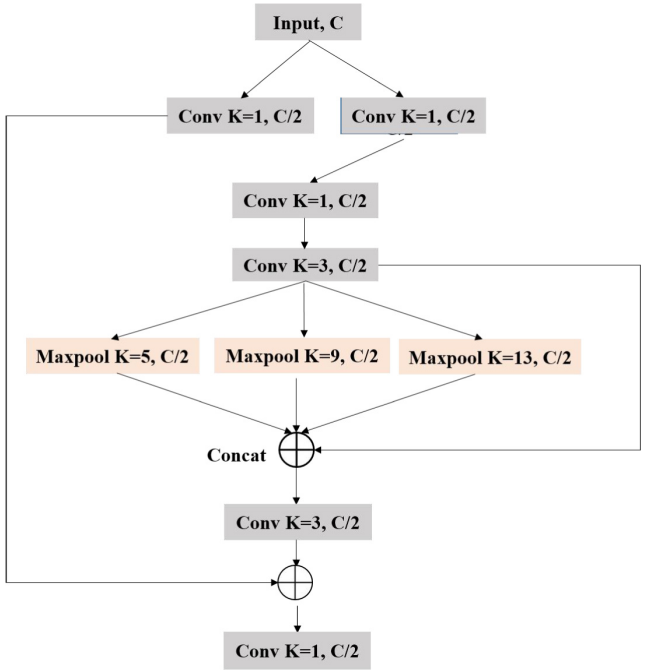


Fig. 5. SPPCSP module structure.

are equally divided into four parts with void rates of 2, 4, 8, and 16 convolution outputs. After the feature is output by the dilated convolution, the output features with dilated rates of 1 and 2 are added first, and then the convolution output feature with a dilated rate of 4 is added again. Finally, feature fusion is performed through the concat operation. This multiscale and multilayer feature fusion strategy accords more with the extraction of SAR image spatial features.

E. Loss

The loss function is used to express the degree of the difference between the prediction and the actual data. A suitable loss function is crucial to training the optimization model. Our training loss function consists of three parts

$$L = \lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{obj}} + \lambda_3 L_{\text{bbox}} \tag{12}$$

where λ_1, λ_2, and λ_3 are the weight coefficients (we set $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 5$); L_{cls}, L_{obj}, and L_{bbox} represent classification loss, target loss, and regression loss, respectively.

The target loss and class loss use the cross-entropy loss function

$$L(p_i, p_i^*) = -\left[p_i \log\left(p_i^*\right) + (1 - p_i) \log\left(1 - p_i^*\right)\right] \tag{13}$$

where p_i represents the true value and p_i^* represents the predicted value.

The regression loss uses the IOU loss function. The IOU loss represents the difference between the predicted frame and the real frame, thereby reflecting the detection effect of the predicted detection frame, and is defined as follows:

$$L = 1 - \text{IOU} \tag{14}$$

where L represents the cross-and-match ratio between the predicted box and the real box.

IV. Experiments

To evaluate the effectiveness of AFSar and the performance of SAR target detection, a series of verification experiments are implemented in this section. This section is divided into three parts. The first part mainly introduces the SSDD dataset and elaborates some settings in our experiment. The second part describes the evaluation indicators used in the experiments performed in this article. The third part evaluates the influence of each module in AFSar on detection performance through ablation experiments and compares the detection performance with that of other target detection algorithms.

A. Dataset and Experimental Setup

To evaluate the performance of our method and comparison with other methods, we use SSDD, the most commonly used dataset in SAR image detection. SSDD is the first publicly available dataset dedicated to ship target detection in SAR images. SSDD was created by Li *et al.* [38]. The images in SSDD are obtained by RadarSat-2, TerraSAR-X, and Sentinel-1 sensors and have four polarization modes: HH, HV, VV, and VH. The resolution is between 1 and 15 m, and the size of each image is between 500 and 600. Between pixels, there are 1160 pictures and 2456 ship targets. Datasets link https://pan.baidu.com/s/1E8ixqK5AVfXc98UgQmpqaw, and the ectraction code is trnt, the decompression code is 12345qwert.

To be consistent with other articles, we randomly divide SSDD into three parts, namely, a training set, validation set,

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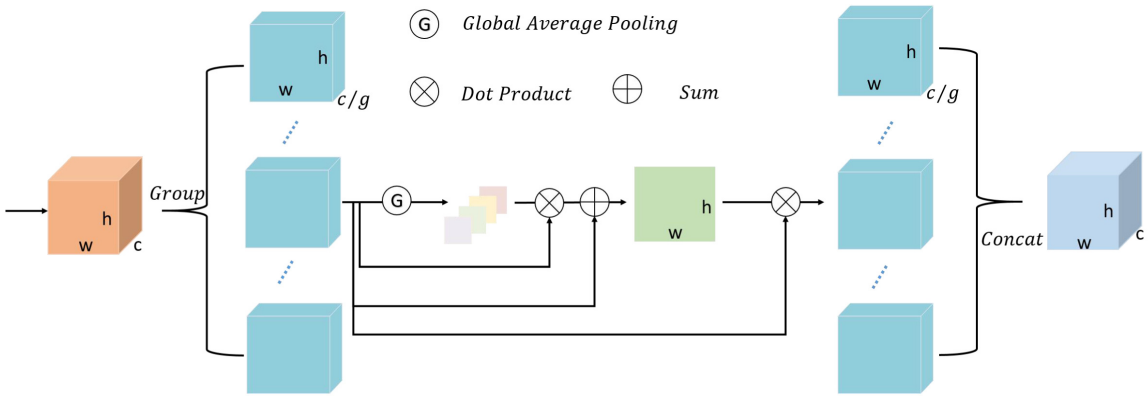


Fig. 6. Spatial attention module structure.

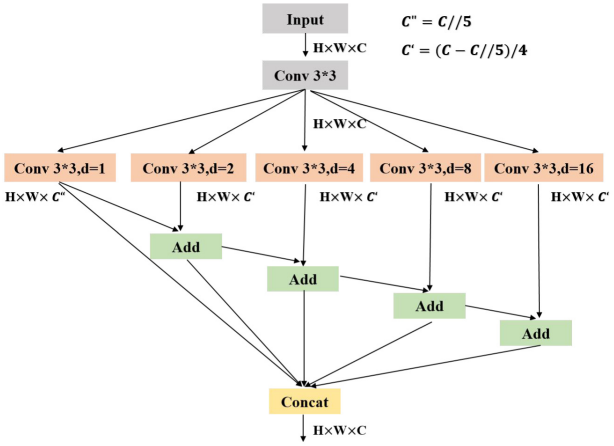


Fig. 7. ESP module structure.

and test set, at a ratio of 7:2:1. The image size of the input model is resized to 640 × 640, and the batch size is set to be 8 by using mosaic, RandomAffine, RandomFlip, and the contrast conversion online enhancement strategy. The optimizer uses SGD; the learning rate is set to be 0.01; the weight decay and momentum are 0.0005 and 0.9, respectively; and 300 epochs are trained. All experiments in this article are performed on an operating system equipped with a GeForce GTX2080Ti and Ubuntu18.0.

B. Evaluation Index

To quantitatively analyze detection performance, we use several of the most commonly used evaluation indicators, such as accuracy, recall rate, $F1$ score ($F1$), and average accuracy (mAP).

Accuracy is the ratio of the correct target to all the identified targets and represents the correct rate of the detected target. Accuracy is defined as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{15}$$

where TP represents the number of positive samples predicted as positive samples (that is, the number of samples where the prediction box and GT IOU greater than threshold) and FP represents the number of negative samples predicted as positive samples (that is, the number of samples whose prediction box and GT IOU less than threshold are regarded as negative samples).

The recall rate represents the ratio of the number of correctly identified positive samples to the number of all positive samples in the test set and is defined as follows:

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{16}$$

where FN represents the number of positive samples predicted to be negative samples.

Since recall and accuracy can only reflect a certain aspect of detection performance. Therefore, the introduction of the index $F1$, which is the harmonic average of the recall and accuracy, can more comprehensively reflect the detection performance. $F1$ is defined as follows:

$$F1 = \frac{2 \times P \times R}{P + R}. \tag{17}$$

mAP represents the average value of each class of AP, that is, the area under the P-R (precision–recall rate) curve, which is defined as follows:

$$\mathrm{mAP} = \int_0^1 P(R) dR. \tag{18}$$

C. Experimental Results and Analysis

1) Ablation Experiment: To verify the effectiveness of each module we designed, we performed four sets of ablation experiments, namely: 1) using baseline YOLOX; 2) keeping other parts of the module unchanged, and replacing the backbone network with MobileNetV2S; 3) keeping other parts of the module unchanged, and then replacing the backbone and PAN with MobileNetV2S and CSEMPAN; and 4) using AFSar.

1) *MobileNetV2S:* In Experiment (2), we first replace darknet (used in Experiment 1), the backbone network of baseline YOLOX, with MobileNetV2S. The performance is shown in Table II. Compared with the mAP obtained by darknet in Experiment (1), the mAP obtained by MobileNetV2S in Experiment 2) dropped by 0.005, but Flops dropped significantly from 21.82 to 8.66 G. The loss of accuracy due to Flops is negligible. Currently, due to the industry's need for algorithms to be transplanted to boards, lightweight

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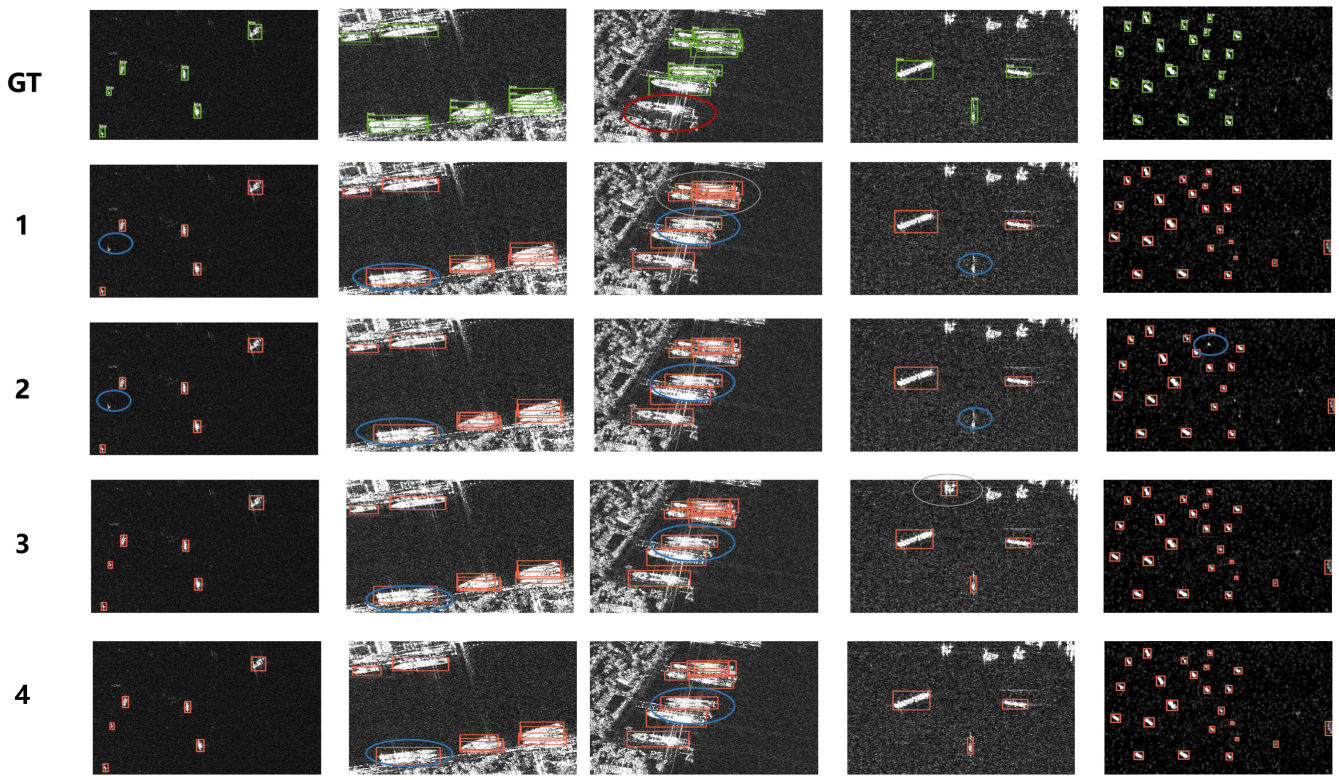


Fig. 8. Visual detection effect of the ablation experiment. The labels 1, 2, 3, and 4 correspond to the four experiments in Table II(1)–(4), respectively. The blue box is the missed detection target, the gray box is the target of misdetection, and the red box is the target that was missed when marked.

TABLE II
AFSAR ABLATION EXPERIMENT

ID	Backbone		Neck		Head		P	R	F1	mAP	Flops
	Darknet	MobileNetV2S	PAN	CSEMPAN	yoloxhead	ESPHead					
(1)	✓		✓		✓		0.932	0.979	0.955	0.972	21.82
(2)		✓	✓		✓		0.912	0.973	0.942	0.967	8.66
(3)		✓		✓	✓		0.937	0.977	0.957	0.975	9.43
(4)		✓		✓		✓	0.941	0.982	0.961	0.977	9.86

models that are limited by board computing power have gradually shown their importance, and lightweight models have more practical significance. In addition, according to the visual detection target in Fig. 8, the detection performance on small targets is slightly worse than the performance of the baseline network with MobileNetV2S as the backbone. As shown in the third row of the fifth column in Fig. 8, YOLOX easily detects dense targets but otherwise causes false detection, as shown in the second row of the third column in Fig. 8.

2) *Attention Enhancement PAN-CSEMPAN:* CSEMPAN aggregates the spatial information of the target by adding the attention module. The experimental effect is shown in Table II(3). Compared with the mAP of YOLOX, the mAP of PAN-CSEMPAN increased by 0.01, and the Flops is reduced by 11.39 G. Compared with Table II(2), Table II(3) shows that mAP increased by 0.008, but Flops increased by only 0.77 G. The increase in the amount of calculation is very small, but the performance is greatly improved, thus showing that our proposed attention-enhanced PAN performs well in SAR image detection. Compared with the second and third rows in Table II, the fourth row in the table shows that the attention-enhanced PAN performs better in detecting small targets and that the number of missed detections is reduced.

3) *ESPHead:* ESPHead obtains the hierarchical information of the target through multiscale cavity convolution to improve detection performance. Table II(4) shows that by adding the ESPHead module, the mAP is improved by 0.02, while the Flops is only 9.86 G, and the detection performance has steadily improved with almost no calculation. The visualization results shown in the fifth row of Fig. 8 show that ESPHead can obtain multiscale spatial information of the target to reduce false detections.

The ablation experiments in Table II verified the performance of each module. In general, each module effectively improved the detection performance or effectively reduced Flops while maintaining the same performance, thereby improving the weight of the model. According to the visualization results

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TABLE III
COMPARISON OF STATE-OF-THE-ART METHODS

	Method	P	R	F1	mAP	Flops(G)
Anchor-based	Yolov3[1]	0.873	0.961	0.915	0.959	77.54
	RetinaNet[4]	0.877	0.948	0.911	0.901	81.69
	SSD300[7]	0.879	0.954	0.915	0.923	154.45
	Faster R-CNN[10]	0.923	0.982	0.952	0.964	70.75
	Cascade R-CNN[13]	0.926	0.980	0.952	0.967	119.05
Anchor-free	FCOS[15]	0.872	0.925	0.898	0.951	78.63
	CenterNet[18]	0.803	0.933	0.863	0.945	20.54
	CornerNet[20]	0.877	0.921	0.898	0.936	707.75
	AFSar	**0.941**	**0.982**	**0.961**	**0.977**	**9.86**

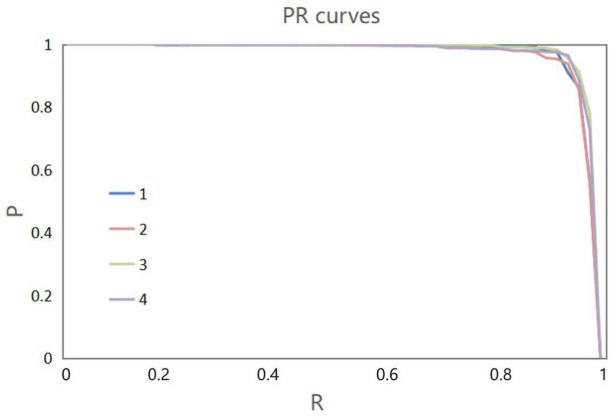


Fig. 9. PR curves 1, 2, 3, and 4 of the ablation experiment represent (1)–(4), respectively, in Table II.

in Fig. 8, compared with the baseline, our model has greatly improved in terms of having fewer false and missed detections. There are missing datasets in the first row and third column, but our model can still detect the target, thus showing that our model has excellent generalization performance. According to the PR curve of Fig. 9, the areas under the purple and green lines are similar, the purple line is slightly higher than the green line, and the blue and the light red lines are slightly lower than the purple and blue lines; these results are consistent with the data in Table II. The effectiveness of each module mentioned in our method improved our model.

2) Comparison With the State-of-the-Art Methods: To further verify our method, we compared it with several other detection algorithms that currently perform best. As shown in Table III, we compared two sets of experiments: experiments using the anchor-based method, and experiments using the anchor-free method. In using the anchor-based method, we compared YOLOv3, RetinaNet, SSD300, Faster R-CNN, and Cascade R-CNN. Table shows that the detection performance of the first-stage method is worse than that of the second-stage method. Cascade R-CNN, which has the highest detection accuracy, has a mAP of 0.967. Compared with the mAP obtained by our method, the mAP obtained by Cascade R-CNN is 0.10 lower; however, the Flops obtained by Cascade R-CNN is 119.05 G, which is much larger than 9.86 G, which is the Flops obtained by our method. Our method performs better in terms of accuracy and computational complexity.

The results visualized in Fig. 10 show that YOLOv3 has missed targets and false detections. RetinaNet performs poorly in detecting dense targets. SSD300 has too much model-fitting ability on the training set due to the multitude of models. There are strong results in detecting much of the background as targets in the test set. Faster R-CNN has missed detections in the case of small targets, while Cascade R-CNN performs poorly in detecting dense targets. According to the visualization and the data in Table III, our method performs better in detecting small and dense targets. The PR curve in Fig. 11 further demonstrates the effectiveness of AFSar's detection performance.

In using the anchor-free method, we compared FCOS, CenterNet, and CornerNet. Table III shows that CenterNet has the highest recall rate at 0.933 and that CornerNet has the highest accuracy at 0.877; these results are lower than those obtained by our method by 0.049 and 0.063, respectively. Our method greatly improves accuracy performance. The recall rate and mAP reached 0.982 and 0.977, respectively, and Flops was less than half of CenterNet's 20.54 G, thus indicating that our method performs well on the SSDD dataset. According to the visual detection results shown in Fig. 12, our method can detect both small targets and complex backgrounds well, thus verifying the effectiveness of our proposed CSEMPAN and ESPHead. The area under the PR curve of Fig. 13 shows that the area enclosed by the PR curve of AFSar is the largest; that is, the mAP is the highest, and the performance is the best.

3) Comparison With SAR Target Detection Algorithms: To further verify the effectiveness of our proposed algorithm, we carried out a further performance comparison with the existing advanced SAR target detection algorithm.

Since most of the SAR target detection algorithms are not open source and the related implementation details are not described, we cannot reproduce the related algorithms to compare and visualize the detection results and draw the PR curve. Therefore, to make the comparison as fair as possible, we compared the best results obtained by current algorithms.

The comparison results are shown in Table IV. AFSar's mAP is 0.005 higher than DSDet's mAP, which is 0.972; AFSar's accuracy is 0.013 higher than FBR-Net's accuracy, which is 0.928; AFSar's recall rate is 0.029 higher than TWC-NE's, which is 0.953; and AFSar's $F1$ score is 0.027 higher

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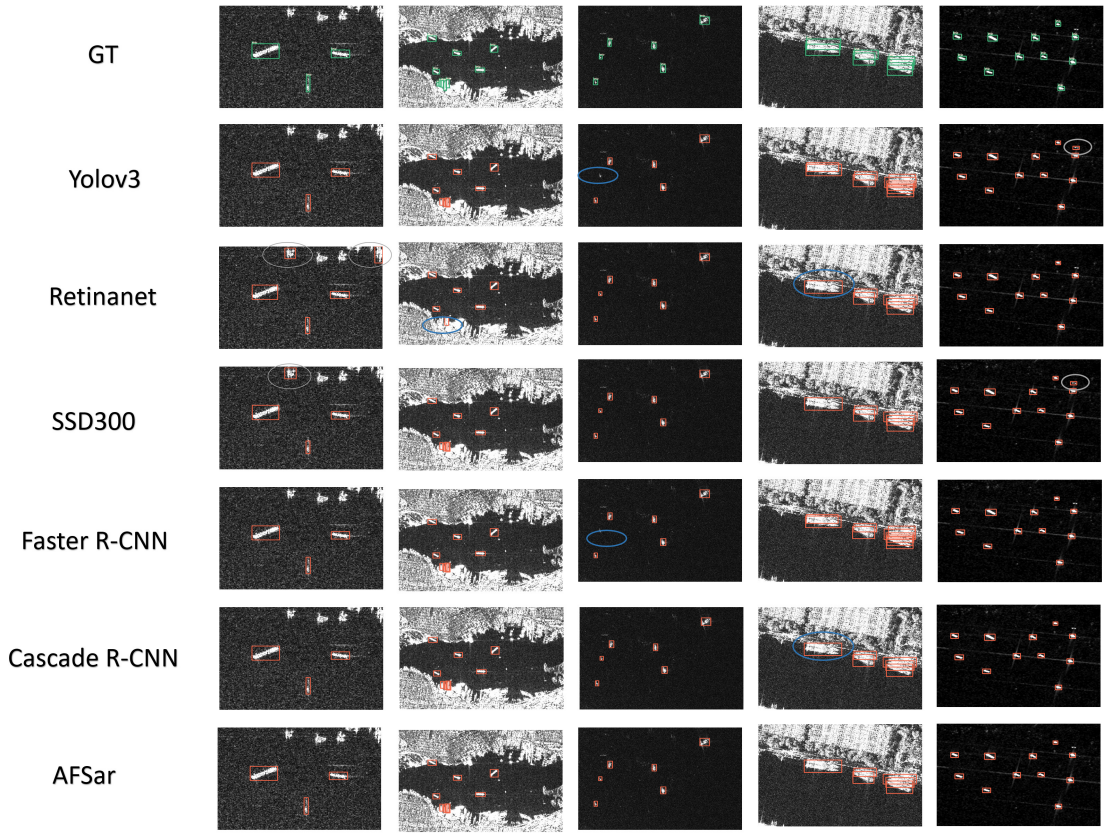


Fig. 10. Visualization of anchor-based target detection algorithm in SSDD dataset.

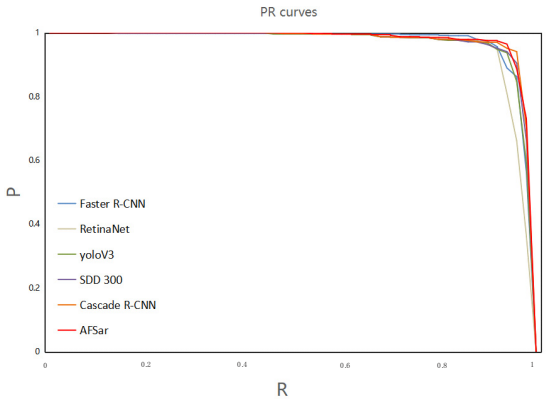


Fig. 11. Comparison of PR curves between AFSar and anchor-based method.

TABLE IV
COMPARISON OF SAR TARGET DETECTION ALGORITHMS

Method	P	R	F1	mAP
FBR-Net[3]	0.928	0.940	0.934	0.941
TWC-Net[6]	0.914	0.953	0.933	-
DSDet[8]	-	-	-	0.972
ISASDNet + r101[11]	-	-	-	0.968
DCMSNN[14]	0.905	0.891	0.898	0.894
CenterNet++[17]	0.833	0.952	0.889	0.951
AFSar	**0.941**	**0.982**	**0.961**	**0.977**

than FBR-Net's, which is 0.934. Therefore, the results of the above multiple indicators show that our AFSar performs best.

4) Comparison With CFAR Detection Algorithms: In addition, we compared the performance of AFsar with representative traditional methods, and the results are shown in Table V. Because the evaluation index of CFAR method and deep learning method are different, we only calculate the scores of P, R, and $F1$ for CFAR method.

TABLE V
COMPARISON OF CFAR DETECTION ALGORITHMS

Method	P	R	F1
CA-CFAR[2]	0.861	0.982	0.917
OS-CFAR[5]	0.840	0.988	0.908
AFSar	**0.941**	**0.982**	**0.961**

CA-CFAR uses the lead-in reference unit between the unit to be detected to detect the SAR image. CA-CAFR first samples the pixels in the window, then calculates the threshold value according to the clutter distribution model, and finally judges whether the pixel belongs to the target according to the threshold value. OS-CFAR first estimates the parameters of the two reference regions, compares the parameter estimation results, selects the larger parameter estimation result as the overall clutter power estimation result, then calculates the threshold T according to the estimation result, and finally judges the pixel whether it belongs to the target.

We tested CA-CFAR and OS-CFAR on all test sets. As shown in Table V, the accuracy of CA-CFAR is 0.861, the recall rate is 0.982, and the $F1$ score was 0.917. The precision of OS-CFAR is 0.840, the recall rate is 0.988, and the $F1$ score is 0.908. Both precisions of CA-CFAR and OS-CFAR are lower than that of AFSar. During the

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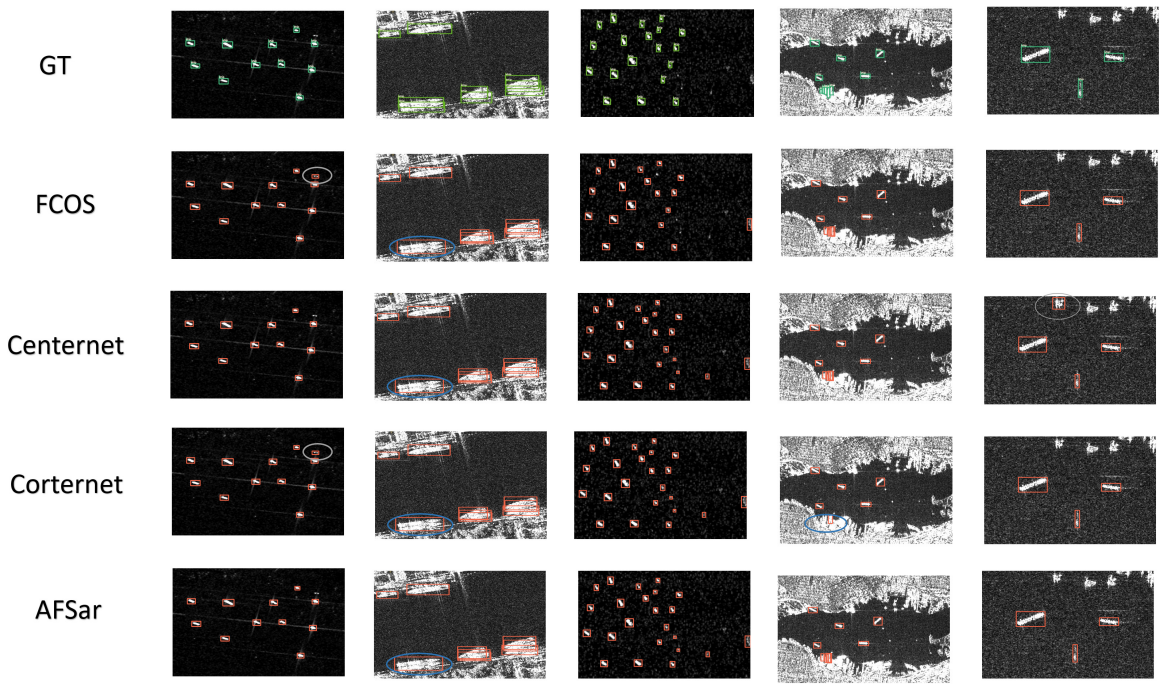


Fig. 12. Visualization of anchor-free target detection algorithm in SSDD dataset.

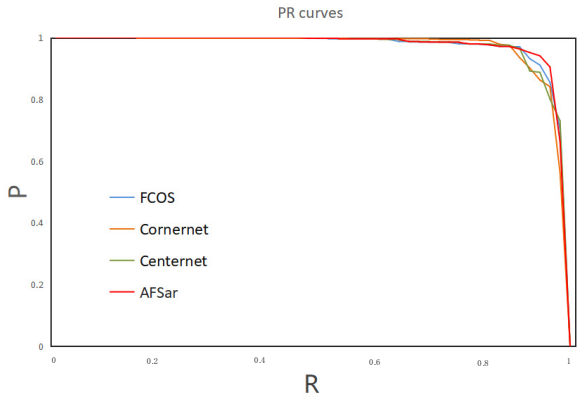


Fig. 13. Comparison of AFSar and PR curves based on anchor-free method.

experiment, we found that the method based on CFAR has better detection effect for small targets and without background clutter images. The effect of small targets is even better than the method based on depth learning. This may be due to the down sampling operation of the method based on deep learning, resulting in the loss of small target information. Therefore, how to combine CFAR and deep learning methods to further improve the detection performance remains to be further studied.

V. Conclusion

SAR image target detection plays a vital role in military and civilian fields. To solve the SAR target detection problems caused by few samples, complex background, strong scattering, sparseness, multiscale, etc., AFSar, an anchor-free SAR target detection algorithm based on multiscale enhancement representation learning, is proposed. Our algorithm consists of three main parts: lightweight backbone-MobileNetV2S, attention enhancement PAN-CSEMPAN, and ESPHead.

The results of a series of ablation experiments on the SSDD dataset verify the effectiveness of each module we designed. The mAP reached 0.977, and the Flops was only 9.86 G; thus, our method reached SOTA in terms of accuracy and computational complexity. Our algorithm can be better deployed to the edge in the future to implement practical applications.

Compared with the baseline YOLOX, our algorithm improved mAP by 0.005 and reduced Flops by 10.96 G. When our algorithm is compared with anchor-based algorithms, the mAP obtained by our algorithm is 0.10 higher than the highest mAP obtained by Cascade R-CNN, and when our algorithm is used, Flops is greatly reduced by nearly 110 G. Compared with other anchor-free algorithms, our algorithm has reached SOTA in terms of accuracy and computational complexity.

In the future, we will conduct more in-depth research mainly from the perspectives of data and algorithms.

1) Data are the key to promoting the application of deep learning, and the current open-source SAR target dataset is small, thus making it difficult to support the rapid development of SAR target detection. To this end, our research group is currently preparing to release a large-scale open-source SAR multiclass target detection dataset, which will be used for studying and verifying SAR target detection algorithms.
2) With the recent development of transformers, we have noticed the great potential of transformers for image processing, but to obtain their benefits, algorithms based on transformers require considerable data. Using our self-built large-scale dataset, we are trying to apply the transformer framework to SAR target detection. Determining how to combine transformers' global field of view and CN's local capture to comprehensively improve SAR target detection performance is our main goal in the next stage.
3) In terms of development, optical image target detection still lags behind SAR target detection. In the future,

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we will devote ourselves to developing a universal SAR image target detection algorithm to satisfy as many application scenarios as possible.

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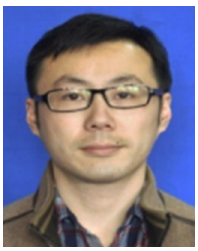
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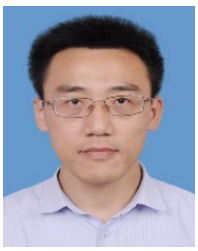
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